PREVIOUS PROJECTS / CASE STUDIES

Selected embedded, IoT, industrial and product-development examples

01 — Smart Soil Testing / NPK IoT Device

Technology: ESP32-S3 • RS485/Modbus • Display • WiFi Web Configuration

- Built a soil measurement platform around an ESP32-S3.
- Integrated RS485/Modbus sensor communication and runtime measurement handling.
- Implemented a local display dashboard for moisture, temperature, conductivity, pH and NPK-related values.
- Added WiFi-based calibration/configuration workflow and web interface concepts.

02 — Industrial / IoT Embedded Product Architecture

Technology: ESP32 / MCU • LTE • Ethernet • WiFi • BLE • Audio / Control

- Designed modular embedded architectures for connected industrial products.
- Considered communication, power management, BMS, LTE and cloud/server connectivity.
- Focused on plug-and-play module replacement for easier service and maintenance.
- Separated hardware, firmware and system-level responsibilities for product development.

03 — STM32 Hardware Security Product Integration

Technology: STM32U5 • SPI • Sensor Integration • Low-level Firmware

- Worked on direct MCU-to-sensor integration requirements for a security-oriented hardware product.
- Focused on transparent interfaces, debugging visibility and vendor documentation.
- Evaluated the need for source-level control and clear low-level communication documentation.

04 — Camera / AI Measurement Concept

Technology: Computer Vision • AI • Industrial Measurement

- Developed a concept for measuring wire-harness branch lengths using camera vision and AI.
- Considered calibration, image processing, geometry and repeatable measurement workflows.
- Targeted high-accuracy automated inspection rather than manual measurement.

Engineering Approach

Projects are approached from the complete embedded-product perspective: requirements, hardware interfaces, firmware/OS architecture, communication protocols, integration, debugging, testing and documentation. The exact implementation is adapted to the customer's MCU/SoC, board, sensors and system requirements.

Information to Start a Project

- Project requirements and expected functionality.
- MCU/SoC/board model and datasheets.
- Schematic/PCB files if available.
- Existing source code or repository if the project is already started.
- Communication protocols and external device details.
- Expected delivery scope and test requirements.

Portfolio Note

Case studies are presented as representative embedded-development work and system concepts. Final implementation, performance and deliverables depend on the customer's hardware, requirements and agreed project scope.